

Wireless Network Protocols

BLE

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Wireless Network Protocols & IoT

- Vital enabler for Internet of Things
 - Vast majority of IoT solutions will include one or more wireless technologies
- Wireless enabled devices/wearables are a major reason for explosion in connected devices.

Global mobile network data traffic:

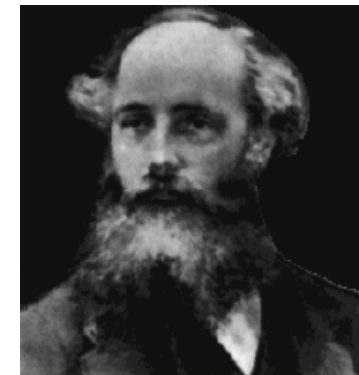
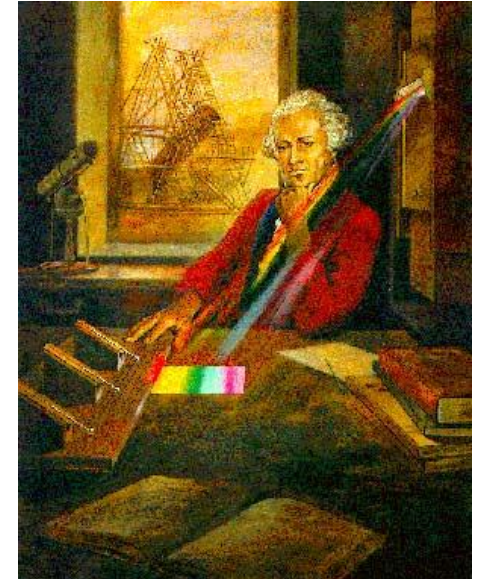
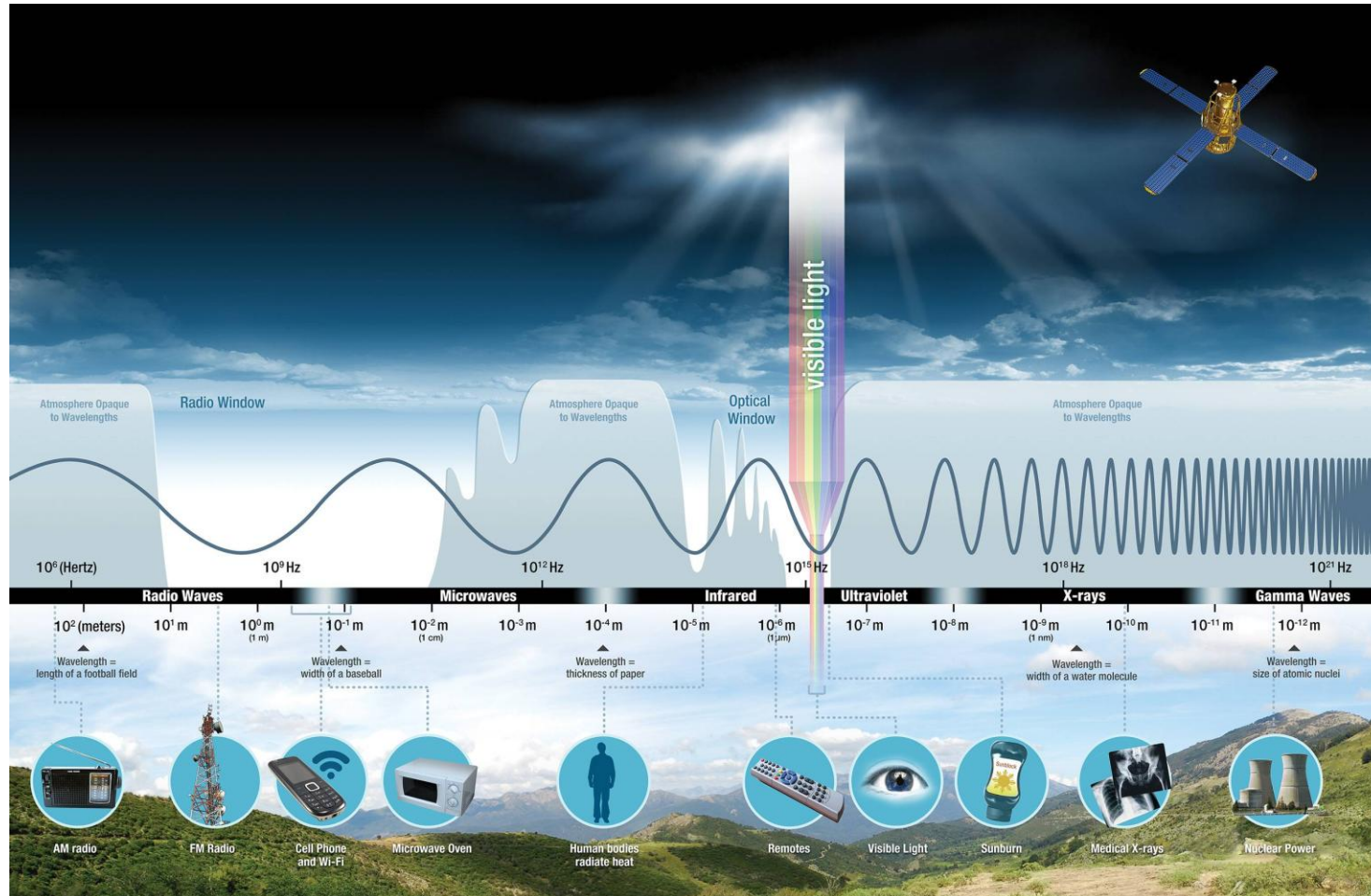
End of 2020: 49 exabytes/month

Forecast for 2026: 237 EB/month

1 EB = 1,000 petabytes (PB) = 1,000,000 terabytes (TB) = 1,000,000,000 gigabytes (GB)



QI – the Electromagnetic Spectrum



https://imagine.gsfc.nasa.gov/science/toolbox/history_multiwavelength1.html

Wireless Advantages

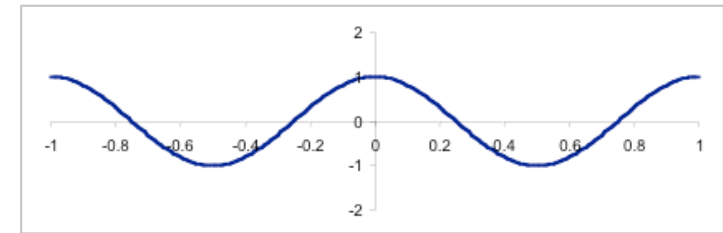
- Look – no wires!
- Mobility
- Deployment
- Productivity
- Convenience



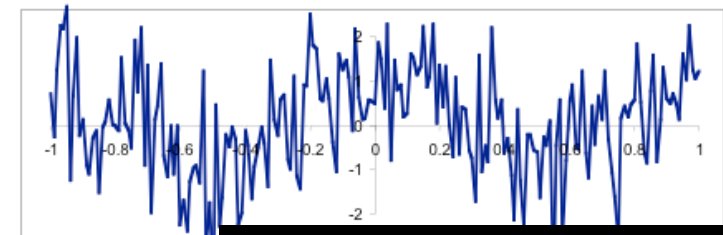
Problems

Physical differences from wired link

- *decreased signal strength*: radio signal attenuates as it propagates through matter (path loss)
- *interference from other sources*: standardized wireless network frequencies (e.g., 2.4 GHz) shared by other devices (e.g., phone); devices (motors) interfere as well
- *multipath propagation*: radio signal reflects off objects ground, arriving at destination at slightly different times
- make communication across (even a point to point) wireless link “difficult” compared to wired link
- Security
- Speed



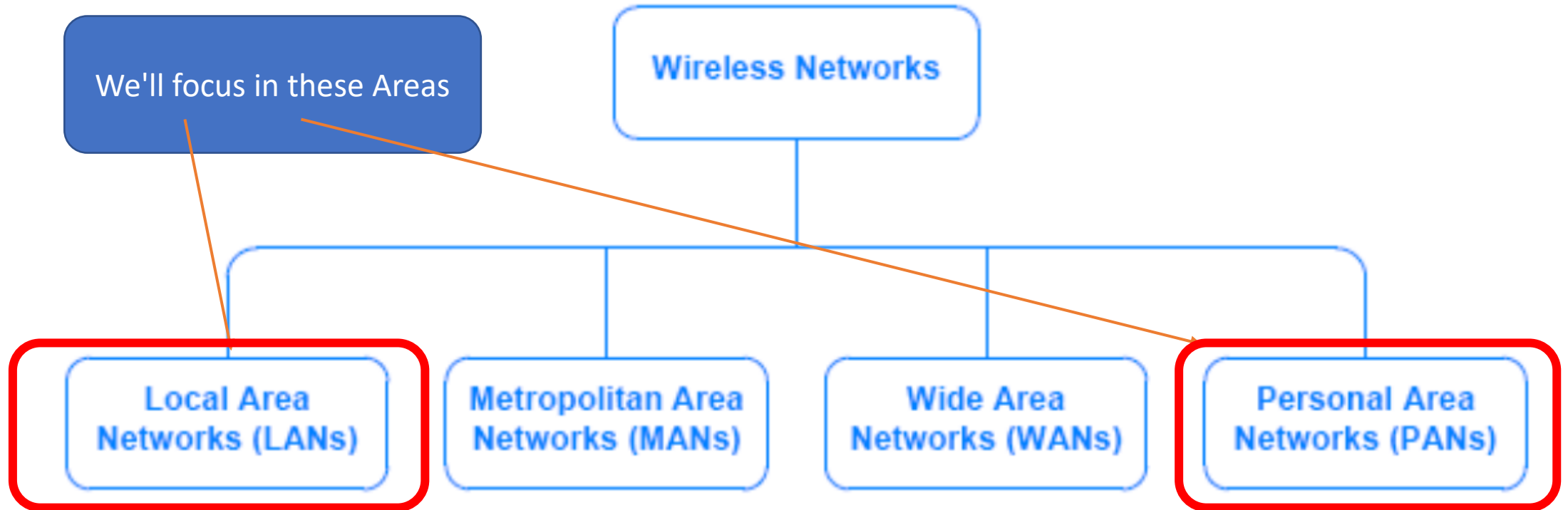
(a)



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Wireless Networks Classification

- Wireless technologies can be classified **broadly** according to network type:



Wireless Technologies

WiFi - family of wireless network protocols, based on the IEEE 802.11 standards (1998)

Bluetooth – IEEE WPAN standard used for device pairing up to 100m distance.

- Bluetooth Low Energy (BLE) – Supports mesh topology to large scale network devices.
- Bluetooth Basic Rate/Enhanced Rate (BR/EDR) – Supports point-to-point topologies and is optimized for audio streaming.

WiMAX (Worldwide Interoperability for Microwave Access) – Alternative broadband wired internet connections. IEEE 802.16 WLAN standard for up 30 miles (50 km).

Cellular Broadband – Carry both voice and data. Used by phones, automobiles, tablets, and laptops.

- Global System of Mobile (GSM) – Internationally recognized
- 3G,4G,5G

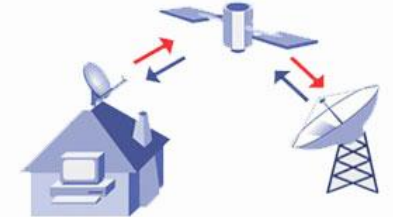
Satellite Broadband – Uses directional satellite dish aligned with satellite in geostationary orbit. Needs clear line of site. Typically used in rural locations where cable/DSL are unavailable.#

Low Power Wide Area Networks(LPWAN), Narrowband (NB) IoT

- long-range communications at a low bandwidth
- Sigfox, LORA, Vodafone NB-IoT



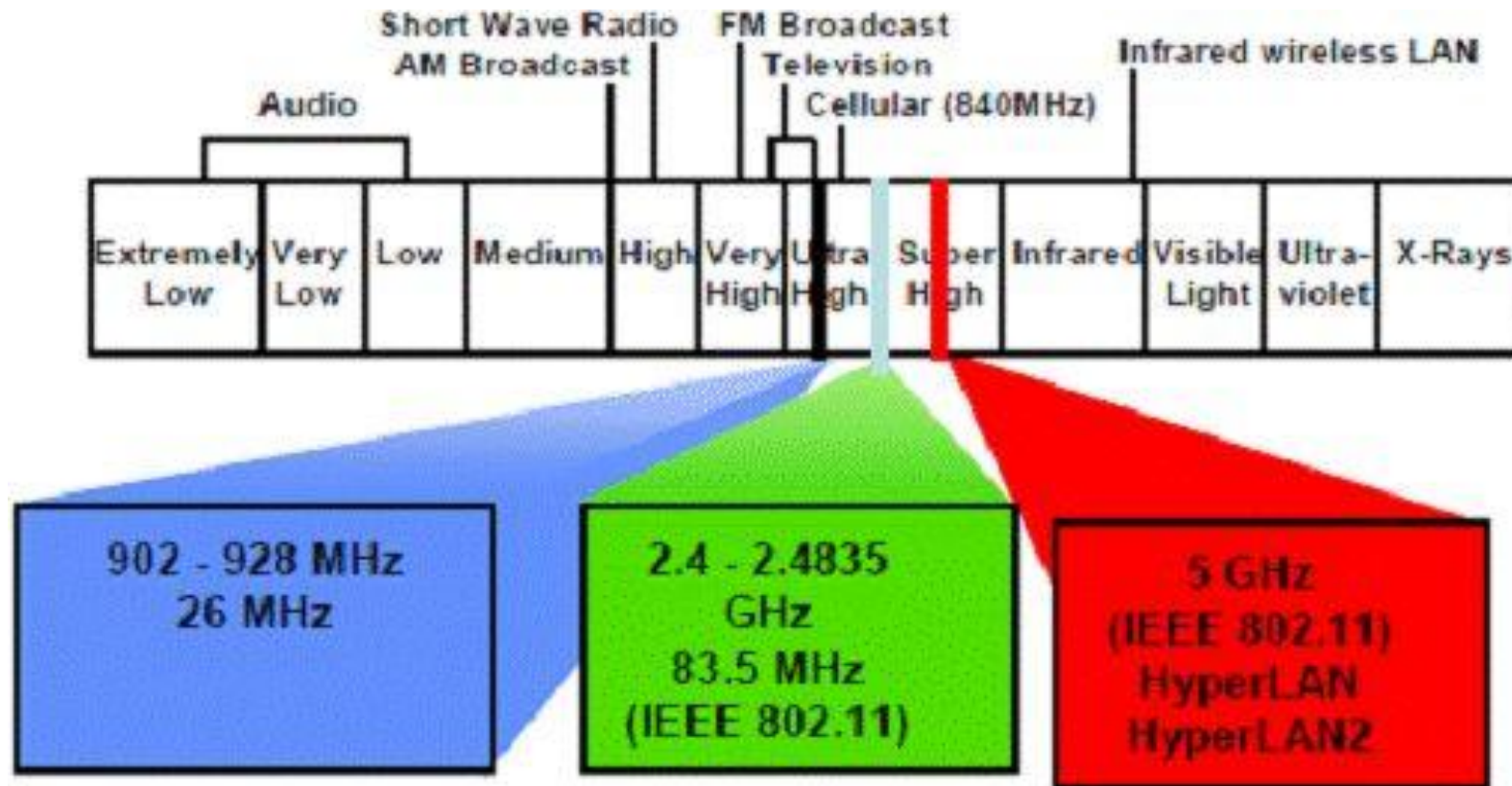
SATELLITE INTERNET



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ISM Wireless Bands

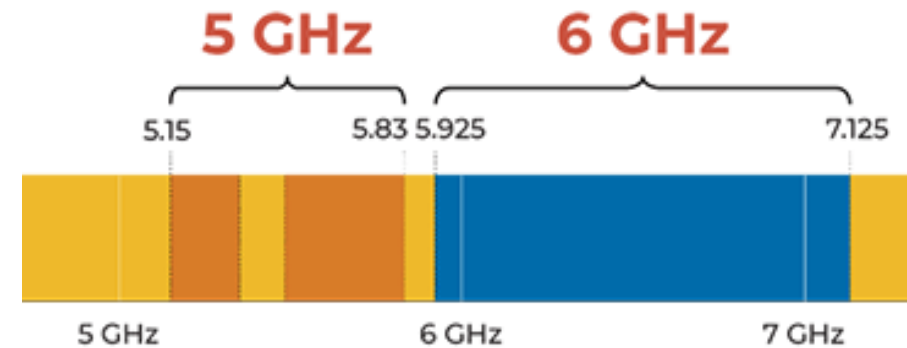
- A region of electromagnetic spectrum is reserved for use by Industrial, Scientific, and Medical (ISM) groups
- These frequencies are not licensed to specific carriers
 - are broadly available for products, and are used for LANs and PANs



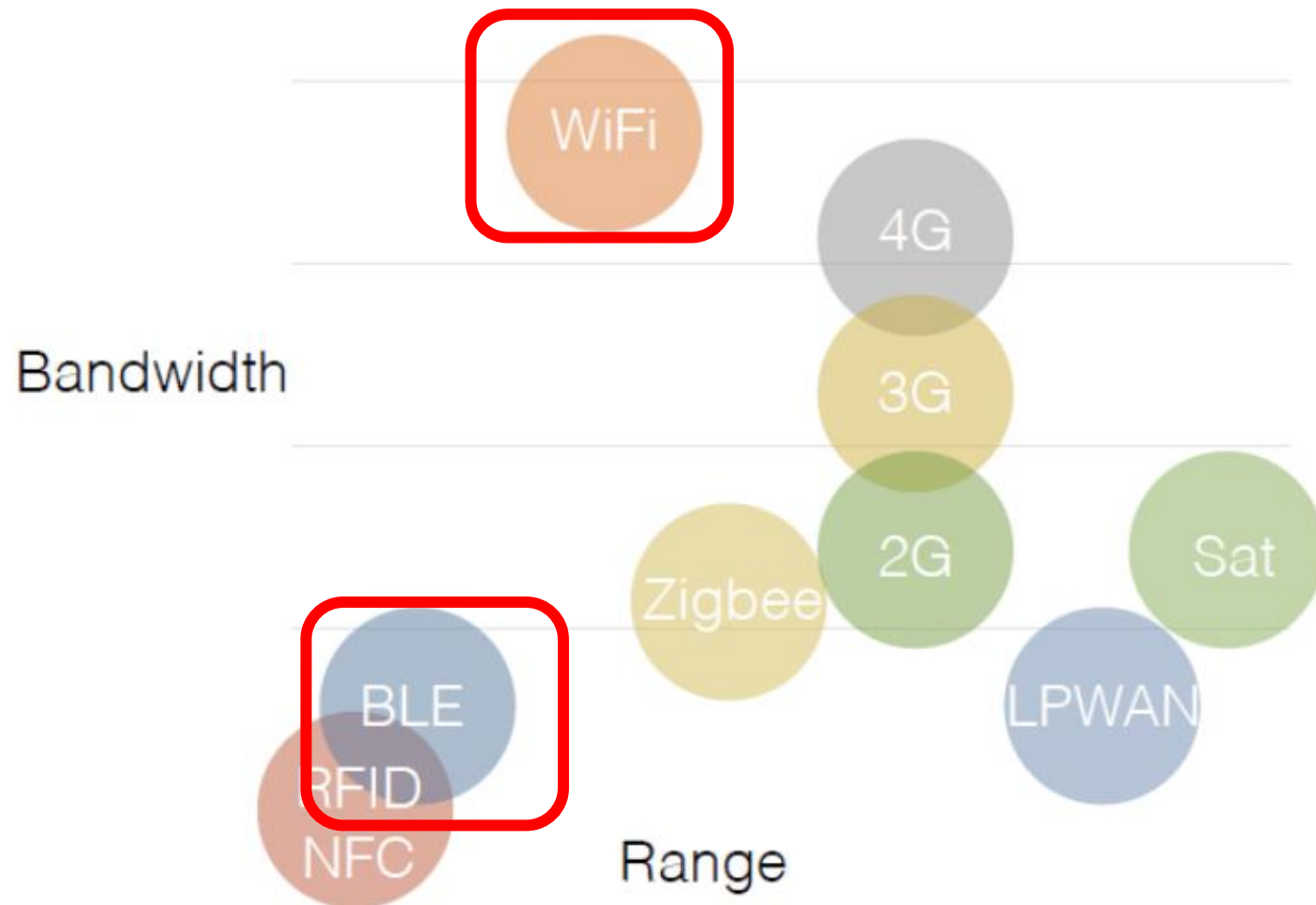
6 GHz Spectrum Allocation

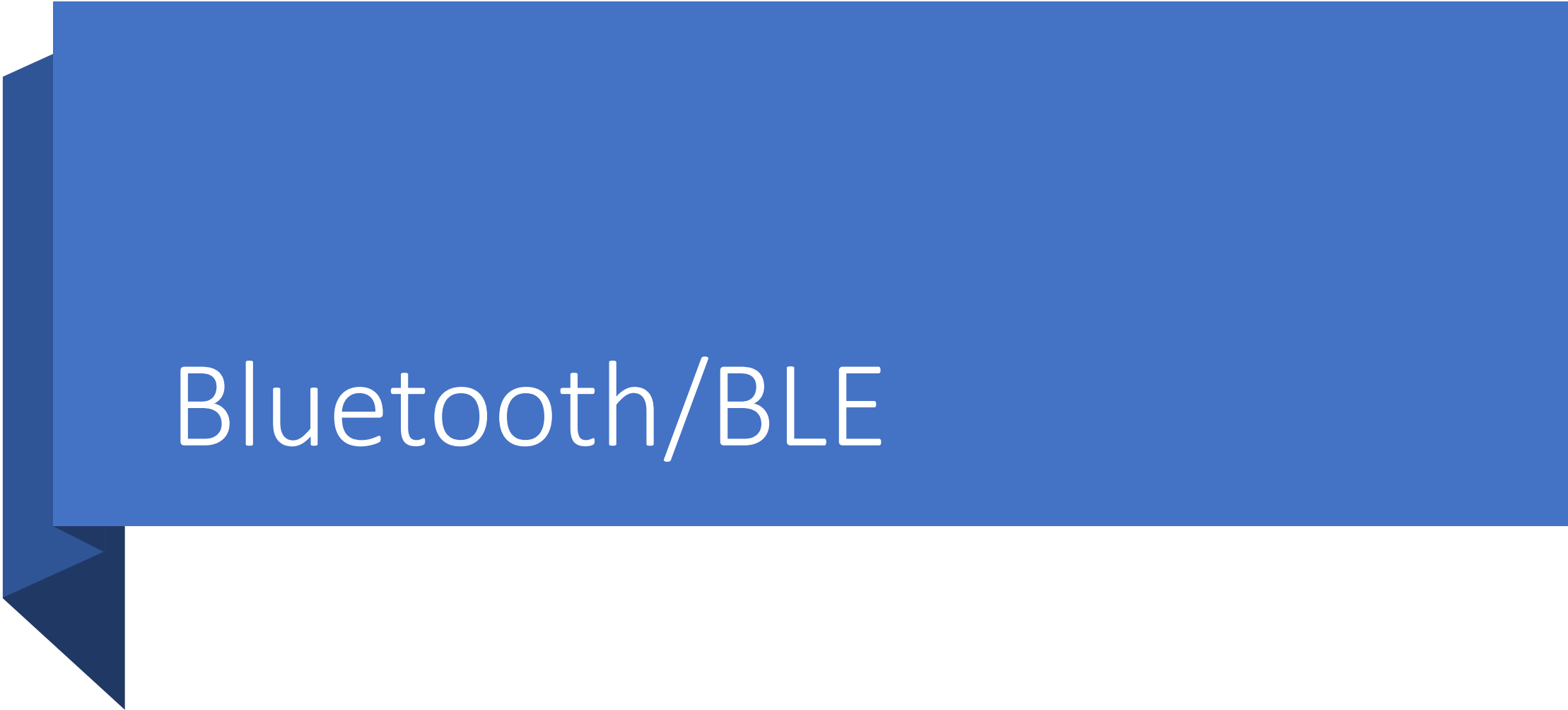
- Traditional ISM bands (2.4 GHz and 5 GHz) are crowded due to widespread use by Wi-Fi, Bluetooth, Zigbee, and other technologies.
- In **2020**: the FCC opened up a part of the 6 GHz range (5.925–7.125 GHz) for unlicensed use, enabling technologies like Wi-Fi 6E.
 - EU has also adopted a similar change
- Not officially part of ISM,
 - New authorised unlicensed spectrum specifically for communication technologies like Wi-Fi
- Advantages:
 - Wider channels – higher speeds possible
 - Low interference

Feature	Description
Frequency Range	5.925–7.125 GHz (varies by region).
Channel Availability	Up to 7 x 160 MHz or 3 x 320 MHz channels.
Licensing	Unlicensed but governed by regional rules.
Primary Use Cases	High-bandwidth, low-latency applications like AR/VR, 8K streaming, and dense IoT ecosystems.



Wireless Technologies Comparison





Bluetooth/BLE

Personal Area Networks (PANs)

- A PAN technology provides communication over a short distance
- It is intended for use with devices that are owned and operated by a single user. For example
 - between a wireless headset and a cell phone
 - between a computer and a nearby wireless mouse or keyboard
- Several standards
 - Frequencies dedicated to Industrial, Scientific and Medical (ISM) band
 - Bluetooth, Near Field Comms(NFC)



Bluetooth

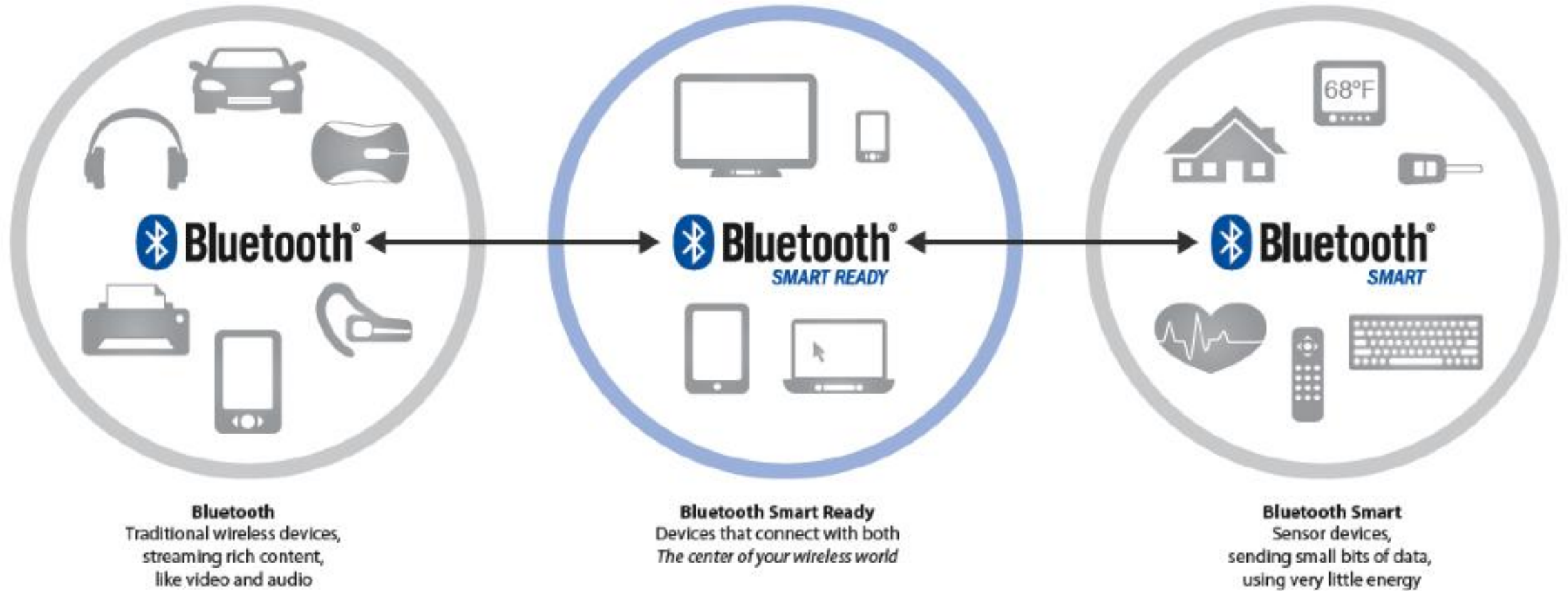
QI: The name "Bluetooth" comes from 10th century Danish king who united Danish tribes into a single kingdom.



- Used for short distance data exchanging, such as Personal Area Network
- RF from 2.4 to 2.485 GHz
- Originally invented by Ericsson (a telecom vendor) in 1994
- Now managed by the Bluetooth Special Interest Group (SIG)
- Frequency hopping spread spectrum (FHSS)
- 16000+ SIG member companies
- Billions of products shipped



Bluetooth Classification



Bluetooth “classic”

- The “conventional” Bluetooth
- Operates in 2.4GHz
- Range: 1m - 100m (10m typical)
- Connection-oriented: audio, file transfer, networking
- Reasonably fast data rate: **2.1** Mbps
- Power consumption:
 < Wifi < 3G,4G,5G



Bluetooth Low Energy

- "Bluetooth Smart"
- Light-weight subset of classic Bluetooth
 - Operates in same freq. Range, 2.4GHz
 - introduced as part of the Bluetooth 4.0 core specification
- Target Apps:
 - Wireless battery-powered sensors eg. heart rate, thermometer, wearables
 - Low bandwidth (you won't be streaming video)
 - Not always on, constrained devices



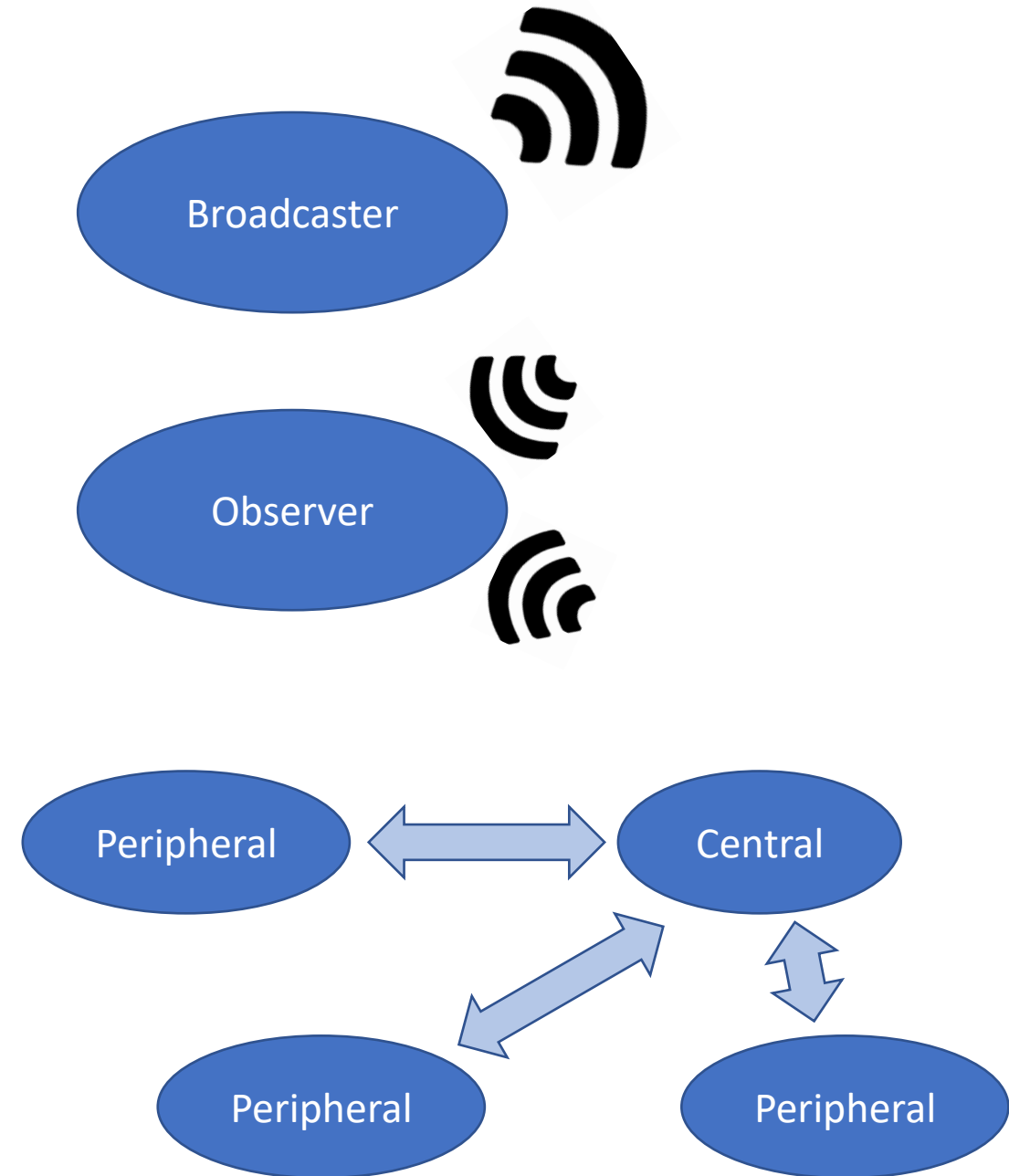
All good for
IoT devices!

BLE vs Classic

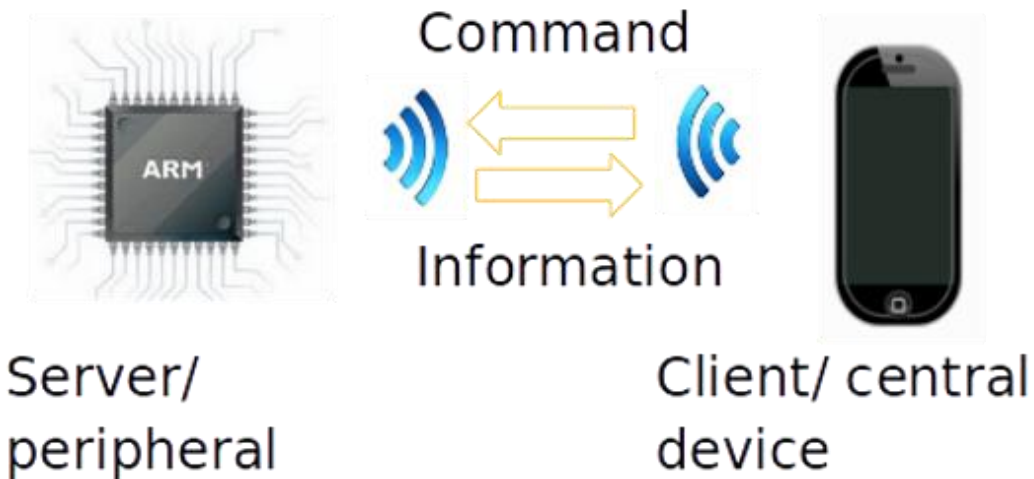
- Bluetooth and Bluetooth Low Energy are used for different purposes
- Bluetooth Classic
 - can handle a lot of data
 - consumes battery quickly
- BLE
 - used for applications that do not need to exchange large amounts of data
 - cheap
 - Marginally further range

BLE Roles

- Broadcaster
 - Transmitter only
- Observer
 - Receiver only
- Peripheral(Slave)
- Central(Master)
 - Can take multiple connections
 - Initiates connection to peripheral
- One Device can have multiple roles



BLE Operation Modes



Peripheral and Central Devices

Master/central : will typically have more computing resources and available energy - a computer or a tablet, for example.

Slave/peripheral : an embedded device - will be constrained in both computing resources and energy.

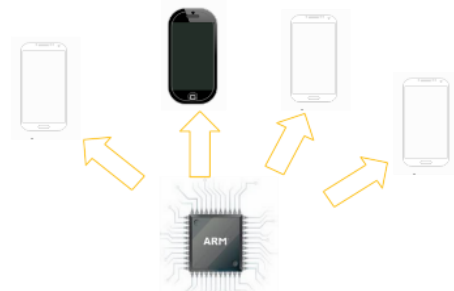
Servers and Clients

Server : the device that has information it wishes to share, and in BLE that is typically the peripheral (the embedded device).

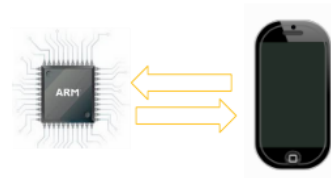
Client : the device that wants information and services, and in BLE that is typically the central device - the phone

BLE Connections

- Initiating Connections
 - The central is free to establish or terminate a connection
 - The peripheral (the BLE device) cannot force the central to scan for BLE devices
- The two modes BLE uses are:
 - Advertising mode : the peripheral sends out Generic Access Profile (GAP) that any device in the area can pick up; this is how central devices know that there are peripherals around.
 - Connected mode : the peripheral and a central device establish a one-to-one conversation. This is how they can exchange complex information.



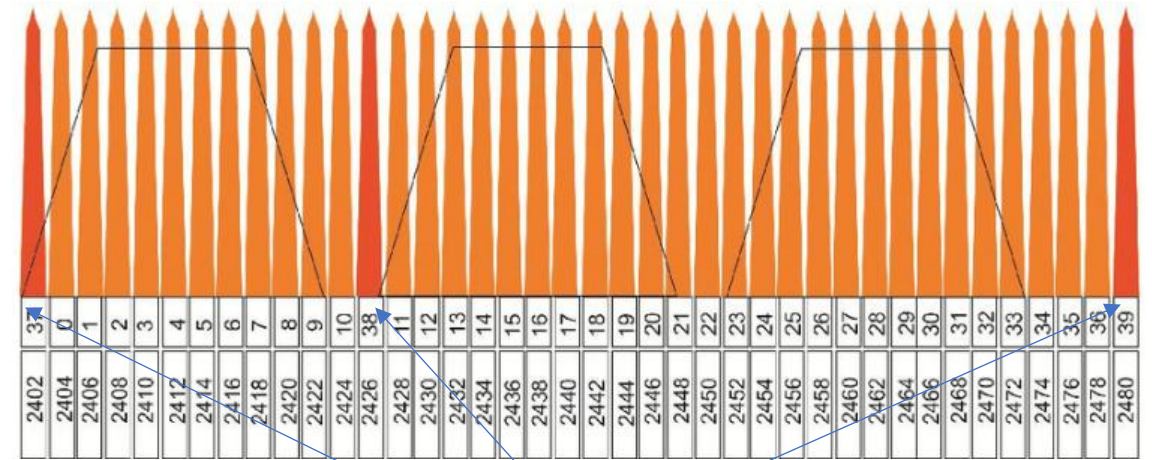
Advertising mode



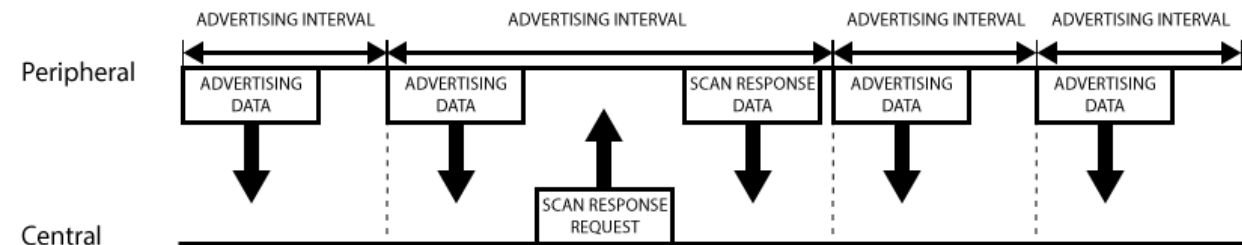
Connected mode

Advertising process

- Operates in 2.4 GHz Band
- Peripheral sets a specific advertising interval and transmit advertising packet
 - longer delays saves power but less responsive
- Transmits on all advertising channels in each interval(channel 37/38/39)
- A listening device can optionally request the scan response payload,
 - peripheral will respond with the additional data.

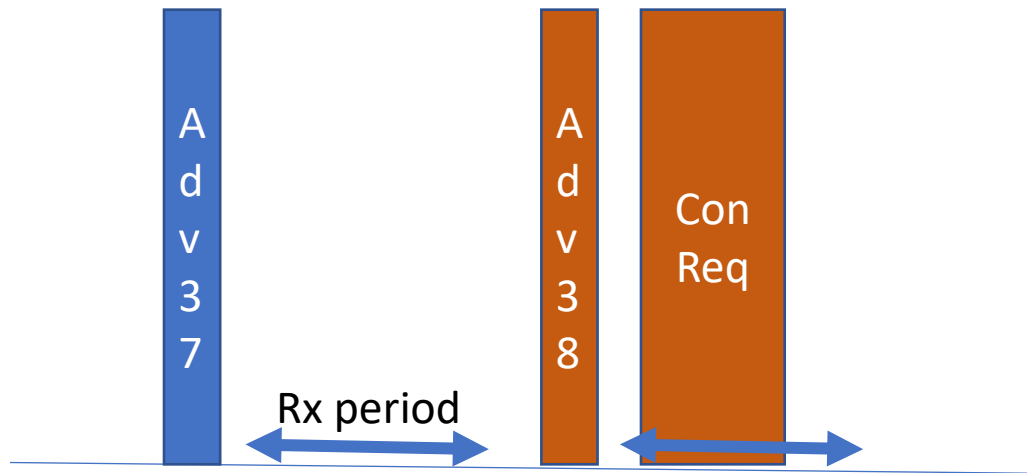


Advertising channels



Connection Process

- Initiated by central device in a specific response period
- Central can issue connection request after receiving advertisement



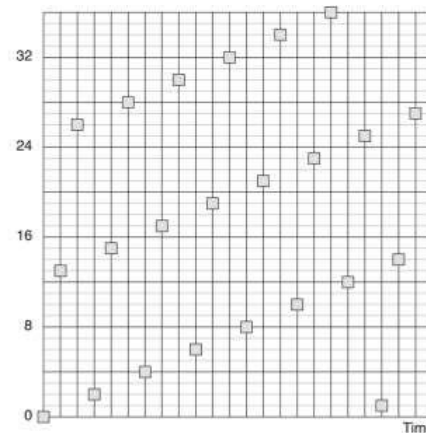
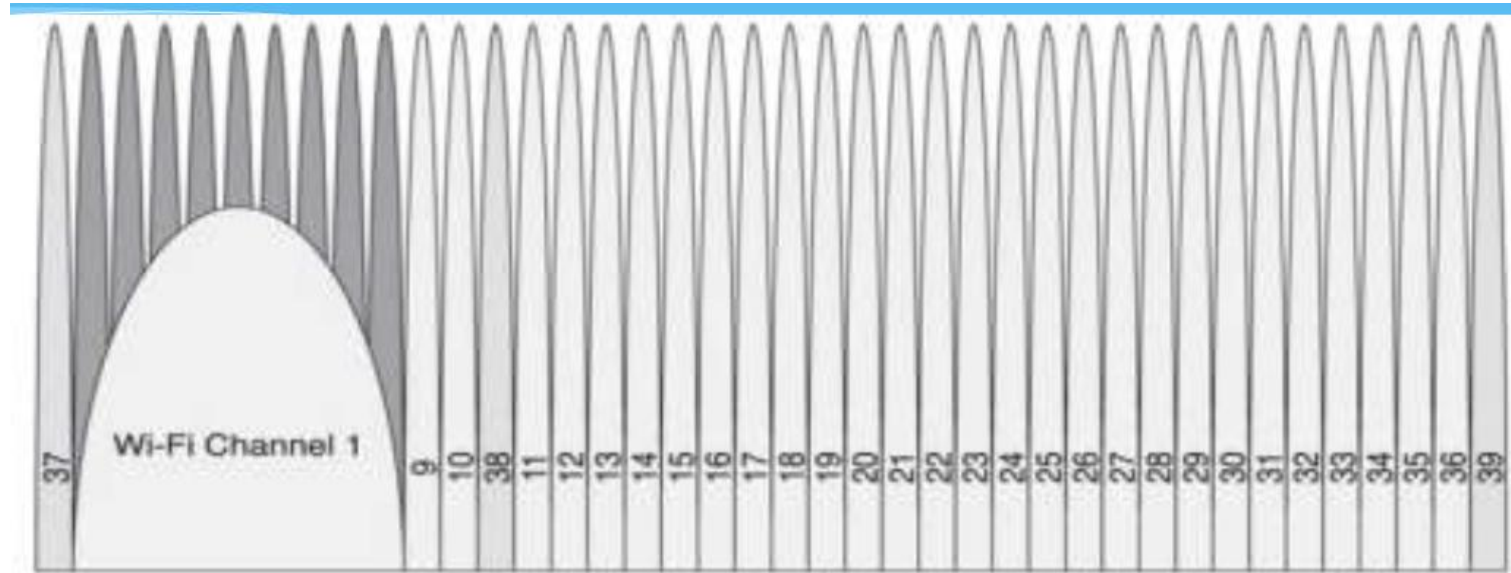
- Peripheral stops advertising after connection request (becomes a slave)
- Slave waits for packets from Master.
 - When received, connection established

Bluetooth LE Device Address

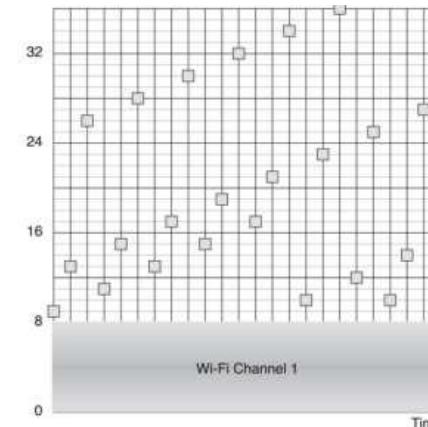
- The Bluetooth ***Device Address*** is a 48-bit (6-byte) number uniquely identifies a device among peers.
 - Similar to an Ethernet Media Access Control (MAC) address
- Think of it as a unique address of a device
- There are two types of device addresses and one or both can be set on a device:
 - Random Address:
 - Random number according to the Bluetooth SIG. Can be static(can only change on power cycle) or Private(resolvable/non-resolvable)
 - Public Address:
 - Calculated just like IEEE Ethernet LAN address

BLE – Adaptive Frequency Hopping

- Once Connected, Channel hopping pattern (from Server) used to minimise interference



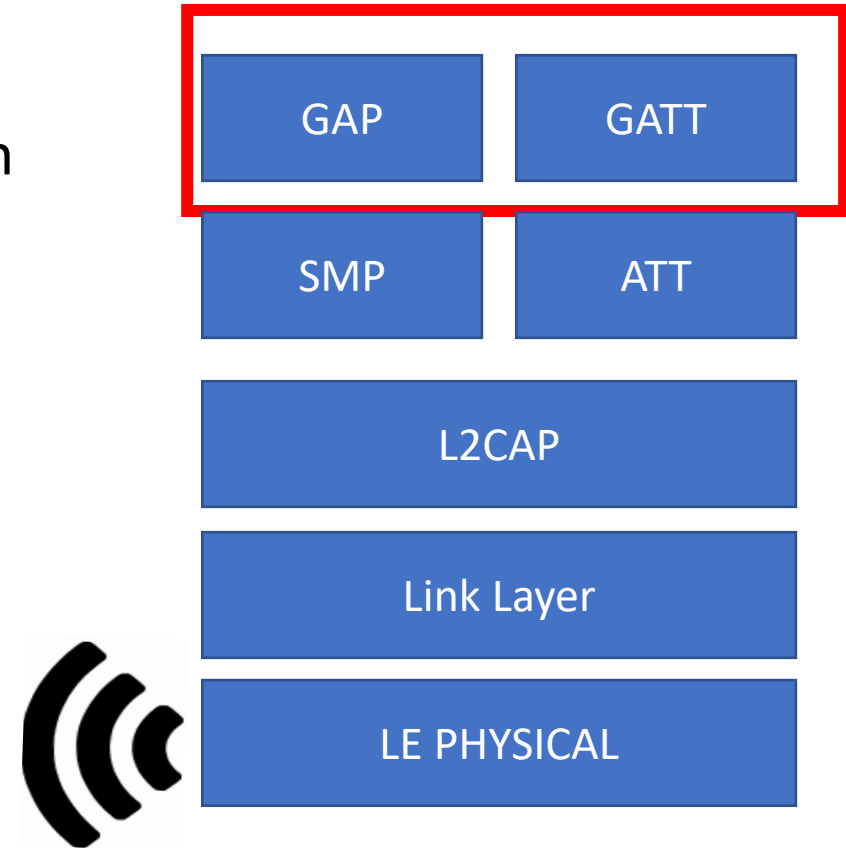
AFH Disabled



AFH Enabled

GAP and GATT for BLE

- Generic Access Profile (GAP) or Advertising
 - Information advertised to central before connection
 - Name of peripheral
 - Is it connectable?
 - Supported features (services)
- Generic Attribute Profile (GATT)
 - How to exchange data once connected
 - Identifies Services, Characteristics and Descriptors

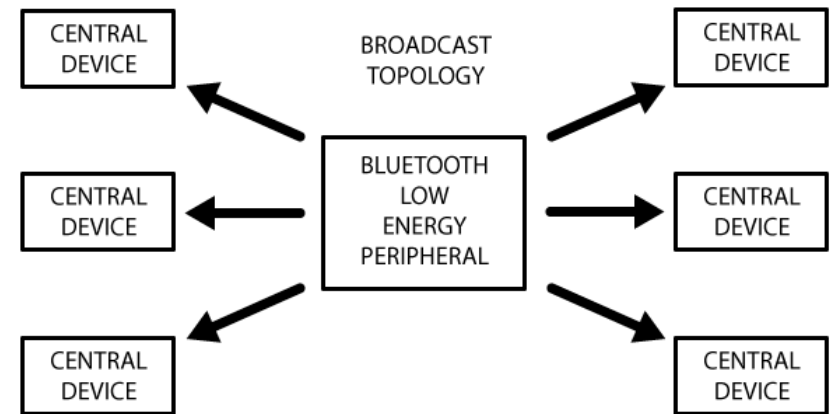


Generic Access Profile (GAP)

- GAP responsible for “device visibility.”
- Determines how two devices can (or can't) interact with each other.
- 2 ways for a device to advertise with GAP
 - *Advertising Data* payload
 - *Scan Response* payload.
- Advertising data payload is mandatory
- Scan response payload is optional
 - allows device designers to fit more information in the advertising payload such as strings for a device name, etc.

Broadcast Network Topology

- Some devices/apps only require advertise data.
 - E.g. app requires peripheral to send data to more than one device at a time.
- Can include small amount of custom data in **31** byte advertising or scan response payloads.
- In this way, BLE peripheral can send data one-way to any devices in listening range
 - BLE Beacons

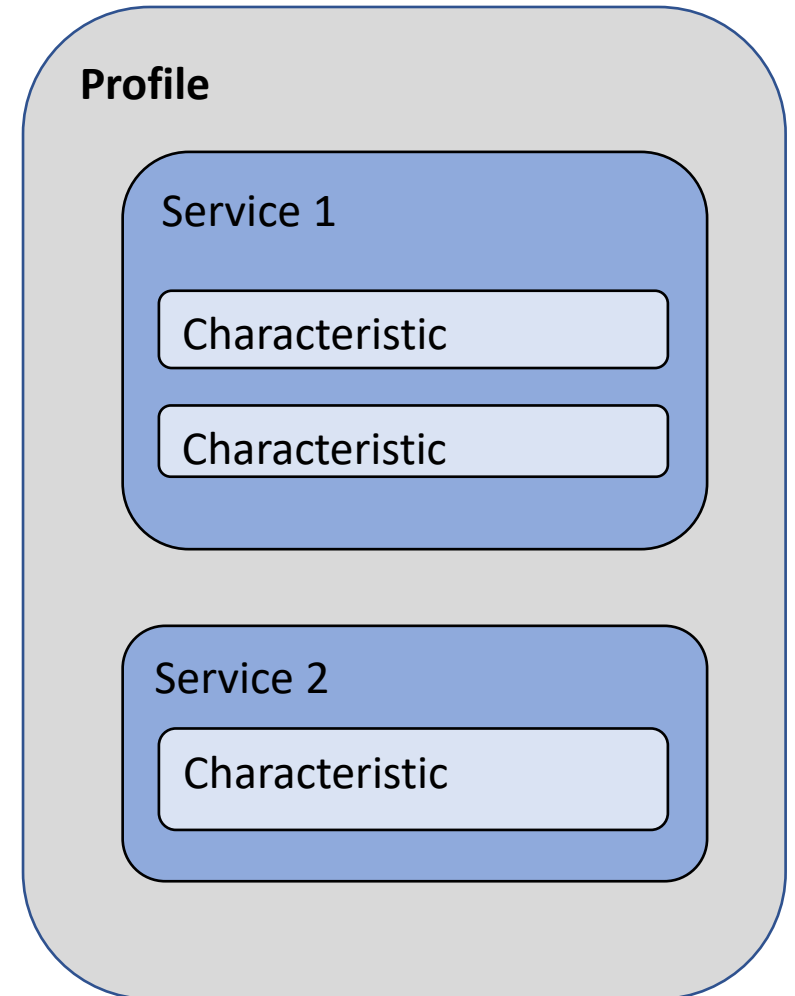


Bluetooth Connection

- Establishing a connection between a peripheral and a central device results in 1 to 1 communication
 - the advertising process will stop
 - no longer be able to send advertising packets
- Communication in both directions
- **Must use GATT services and characteristics to communicate**

GATT Services

- Breaks data up into logic entities
- Contains one or more **characteristics**
- Each service distinguished by unique numeric ID called a UUID
 - 16 bit
- Set of officially adopted BLE services can be seen on the [Services](#)
- E.g. official Heart rate service
 - service has a 16-bit UUID of 0x180D
 - contains up to 3 characteristic
 - ***Heart Rate Measurement, Body Sensor Location*** and ***Heart Rate Control Point***.



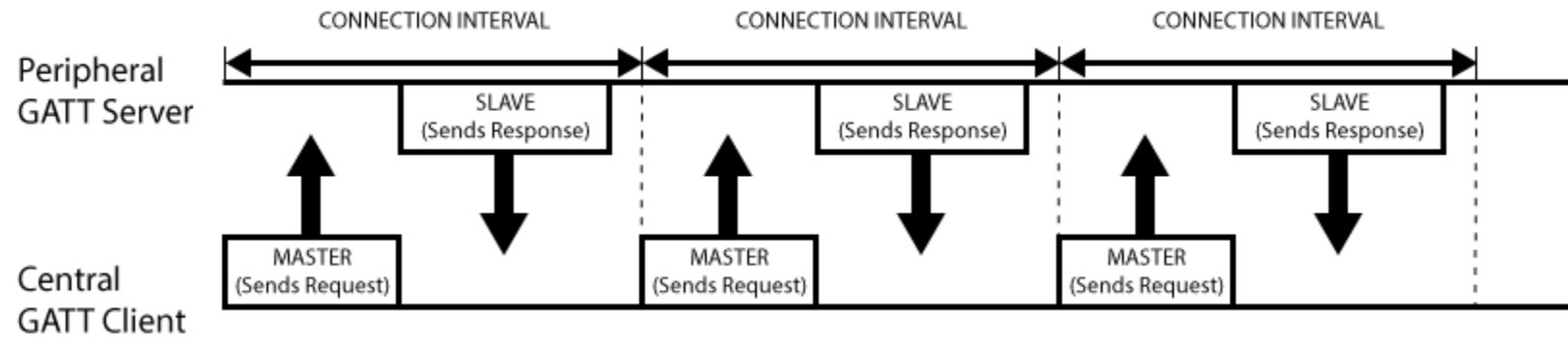
GATT Characteristics

- Represents a single data point
- Similarly to Services, each Characteristic distinguishes itself via a pre-defined UUID
 - Also use [standard characteristics defined by the Bluetooth SIG](#)
- E.G Heart Rate:
 - the [Heart Rate Measurement characteristic](#) is mandatory for the Heart Rate Service
 - Heart rate measurement has UUID of 0x2A37
- If you write apps that use BLE, **characteristics** are what you will be after with your BLE peripheral

BLE UUIDs

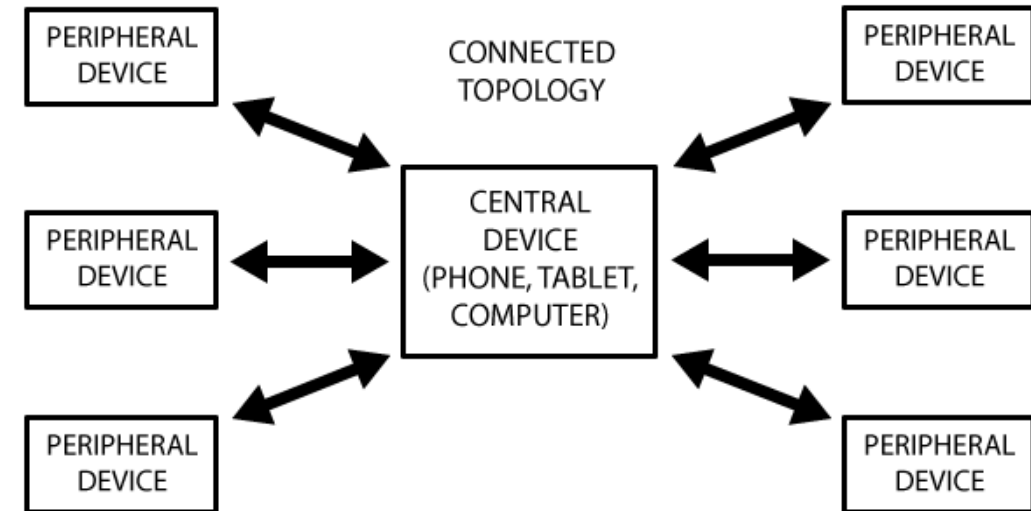
- BLE characteristics are identified by 128-bit UUIDs, written as 32 hex digits in 5 groups: (xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx).
- All standard GATT types use the base: 0000xxxx-0000-1000-8000-00805f9b34fb.
- The Heart Rate Measurement characteristic uses the 16-bit ID **0x2A37** defined by the Bluetooth SIG.
- Substitution into the base UUID: The 16-bit value 0x2A37 is inserted into xxxx → 0000**2a37**-0000-1000-8000-00805f9b34fb.
- Standard vs custom characteristics: Standard characteristics (like HR) use this base+16-bit scheme; custom/vendor characteristics use arbitrary 128-bit UUIDs outside this base.

GATT Transactions



BLE Connected Network Topology

- A peripheral can only be connected to one central device
- Communication is **2** -way
- Central device can be connected to multiple peripherals.
- If data needs to be exchanged between two peripherals, a custom messaging system will need to be implemented
 - all messages pass through the central device.



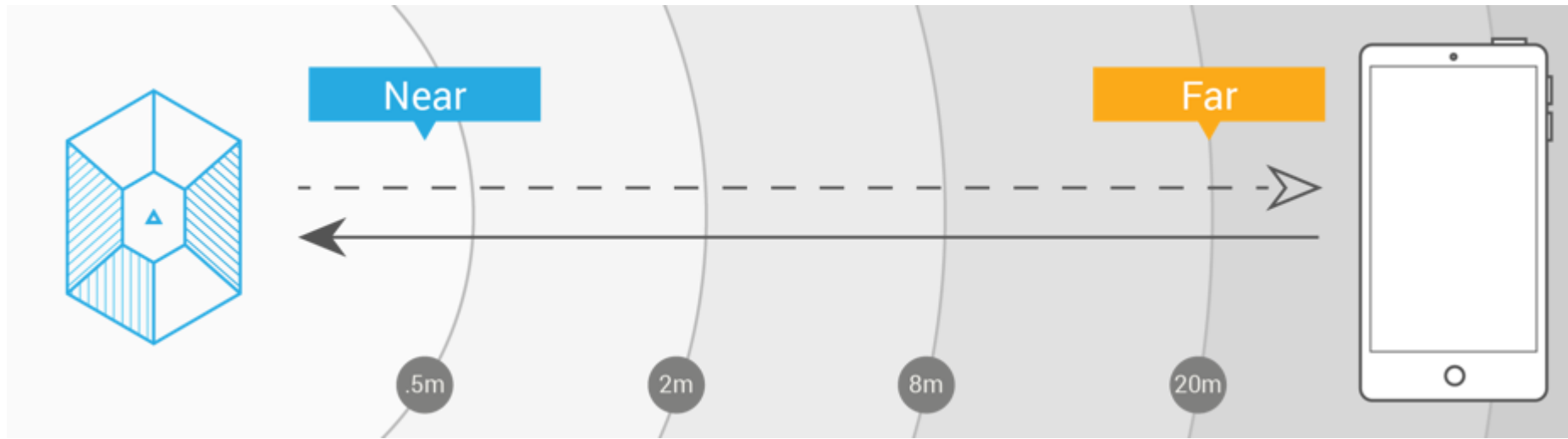


What are beacons?

- Low-cost, low-power transmitters equipped with Bluetooth Low Energy
- Helps mobile devices detect proximity and determine micro-location

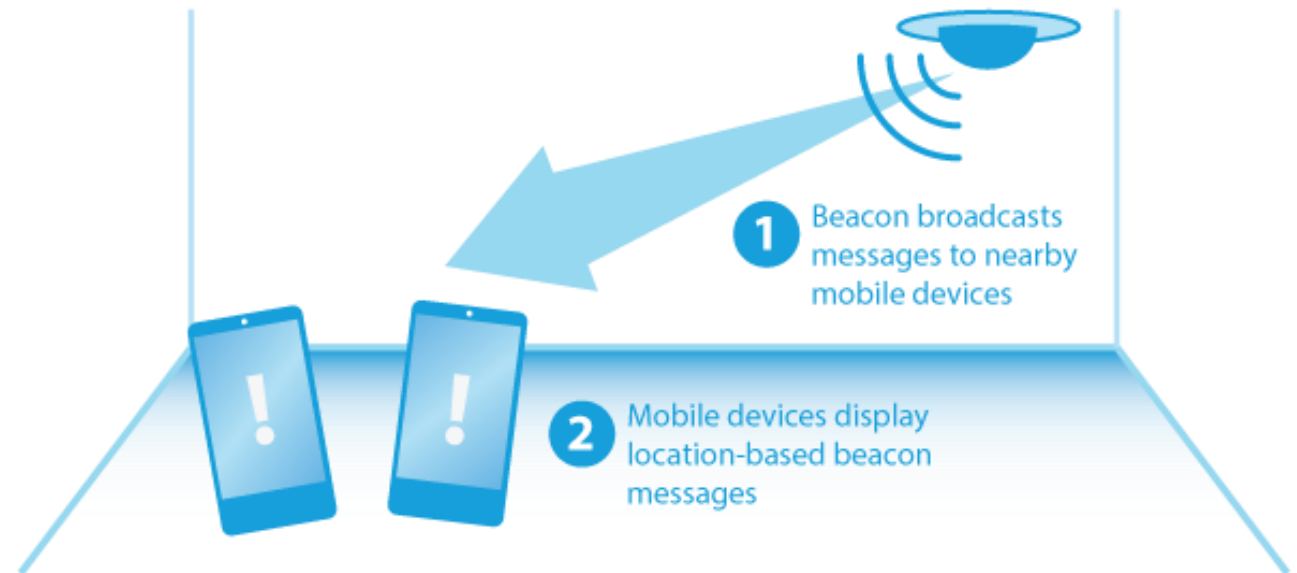
BLE Beacons – How they work

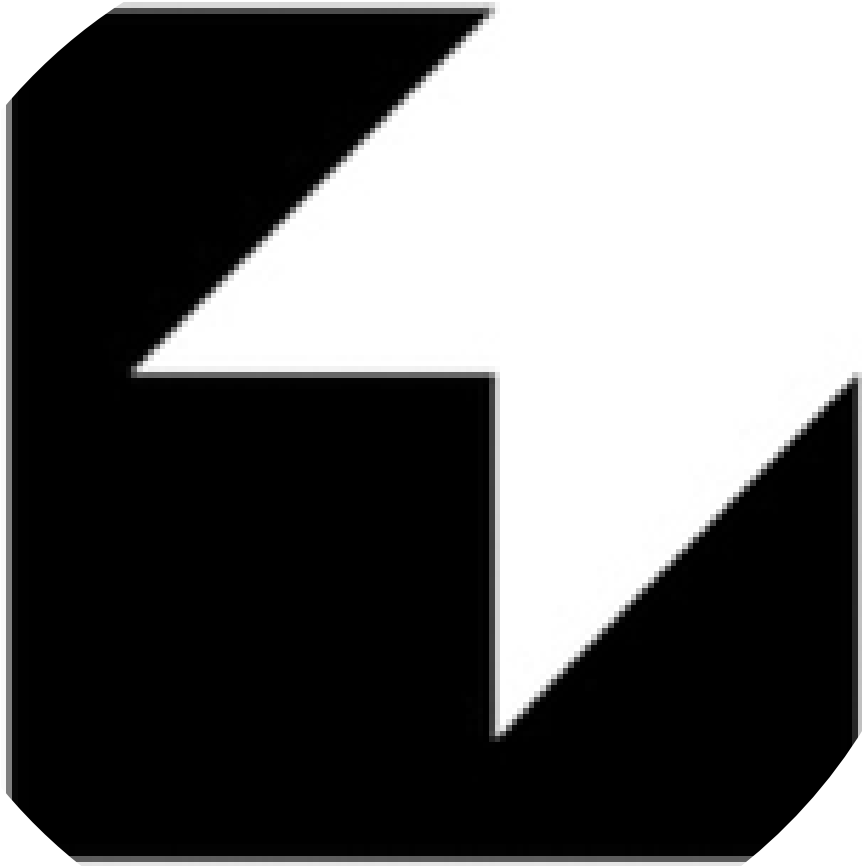
- Beacons don't transmit content, they transmit a series of numbers
- The transmitted signal allows another device to identify it and determine the device's proximity to the beacon



Beacons

- Use Cases
 - Proximity marketing
 - Hyperlocal check-in
 - Retargeting Ads
 - Asset tracking
 - Indoor navigation



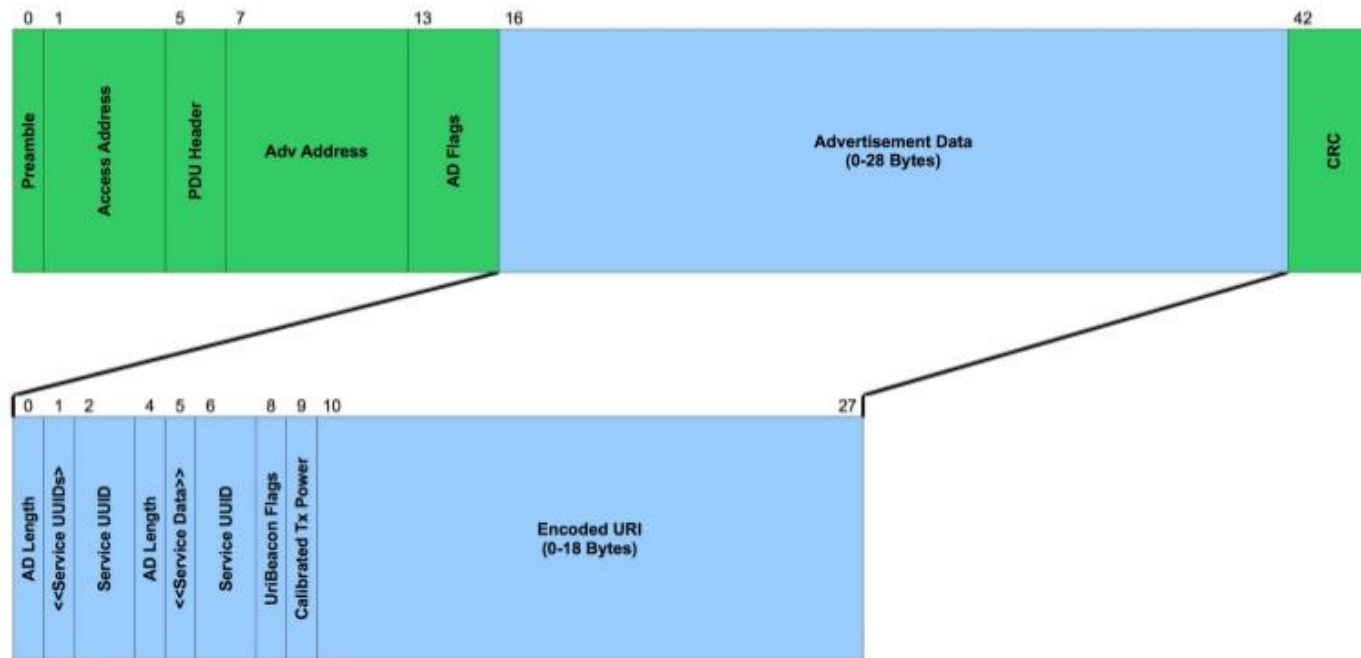


What is Eddystone?

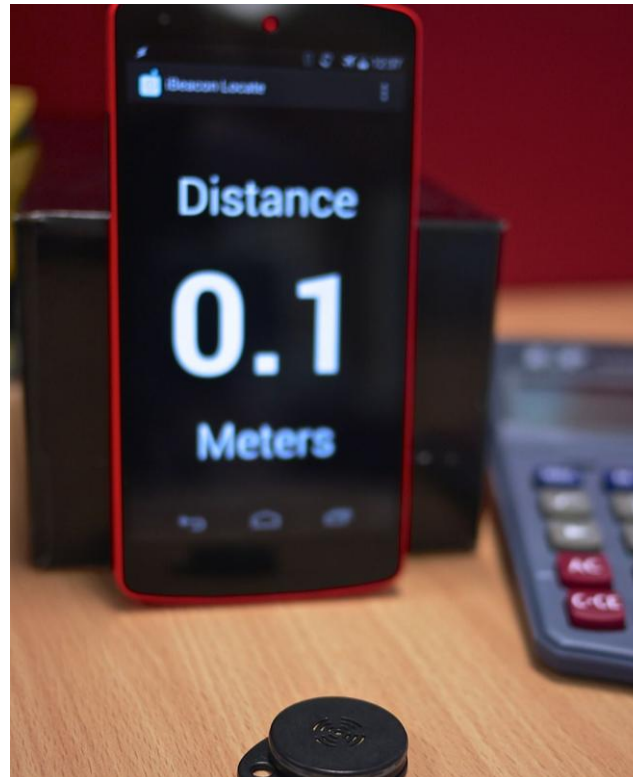
- BLE beacon broadcast standard developed by Google
- Cross-platform and capable of supporting Android, iOS and any platform that supports BLE beacons

Eddystone BLE Advertisement PDU

Technical



What is iBeacon



- Apple's beacon standard
- Android and iOS compatible, native only for iOS
- Broadcasts one advertising packet and a unique ID
- Beaconstac SDK can wake up Android apps, even if they are closed

BLE Power consumption

Chipset	Advertising Interval	Est. Battery Life CR2032	Est. Battery Life CR2045	Est. Battery Life CR2477
Gimbal	100ms	n/a	n/a	n/a
Gimbal	645ms	1 month	2.5 months	4.1 months
Gimbal	900ms	n/a	n/a	n/a
Nordic Semiconductors	100ms	1.2 months	3.1 months	5.1 months
Nordic Semiconductors	645ms	7.0 months	18.19 month	29.3 months
Nordic Semiconductors	900ms	11.1 months	28.7 months	46.29 months
Bluegiga	100ms	0.9 months	2.4 months	3.8 months
Bluegiga	645ms	5.9 months	15.4 months	24.8 months
Bluegiga	900ms	9.3 months	23.9 months	38.5 months
Texas Instruments	100ms	0.7 months	1.8 months	2.9 months
Texas Instruments	645ms	4.1 months	10.6 months	17.1 months
Texas Instruments	900ms	5.6 months	14.4 months	23.1 months